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In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document specifies the mapping:

- performed by the AMF between HTTP responses (Status Codes and Protocol or Application Errors) returned by 5GC NFs to the AMF and 5GMM Cause values sent to UEs;
- performed by the SMF between HTTP responses (Status Codes and Protocol or Application Errors) returned by 5GC NFs to SMF and 5GSM Cause values sent to UEs;
- performed by the SMF between N4 errors returned by the UPF and 5GSM Cause values sent to UEs.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 29.244: "Interface between the Control Plane and the User Plane Nodes".

[3] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".

[4] 3GPP TS 29.531: "5G System; Network Slice Selection Services; Stage 3".

[5] 3GPP TS 29.503: "5G System; Unified Data Management Services; Stage 3".

[6] 3GPP TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".

[7] 3GPP TS 29.509: "5G System; Authentication Server Services; Stage 3".

[8] 3GPP TS 29.502: "5G System; Session Management Services; Stage 3".

[9] 3GPP TS 29.508: "5G System, Session Management Event Exposure Service; Stage 3".

[10] 3GPP TS 29.540: "5G System; SMS Services; Stage3 ".

[11] 3GPP TS 29.511: "5G System; Equipment Identity Register Services; Stage 3".

[12] 3GPP TS 29.507: "5G System; Access and Mobility Policy Control Service; Stage 3".

[13] 3GPP TS 29.525: "5G System; UE Policy Control Service; Stage 3".

[xx] 3GPP TS 29.536: "5G System; Network Slice Admission Control Services; Stage 3".

# 3 Definitions, symbols and abbreviations

## 3.1 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

| | |
|---|---|
| 5GC | 5G Core Network |
| AMF | Access and Mobility Management Function |
| HTTP | Hypertext Transfer Protocol |
| NF | Network Function |
| NSSF | Network Slice Selection Function |
| PCF | Policy Control Function |
| SMF | Session Management Function |
| UDM | Unified Data Management |

# 4 Mapping between 5GC interfaces causes and 5GMM Cause Codes by AMF

## 4.1 General

This clause defines the mapping performed by the AMF between HTTP responses (status codes and Protocol or Application Errors) returned by NF services contacted by the AMF and the5GMM cause values sent to UEs.

No mapping is required for:

- the Nsmsf_SMService_Activate, Nsmsf_SMService_Deactivate and Nsmsf_SMService_UplinkSMS operations described in 3GPP TS 29.540 [10].
- the Npcf_AMPolicyControl service operations described in 3GPP TS 29.507 [12].
- the Npcf_UEPolicyControl service operations described in 3GPP TS 29.525 [13].

## 4.2 Mapping between Nausf services causes and 5GMM causes

### 4.2.1 General

This clause defines the mapping performed by the AMF between HTTP responses (status codes and Protocol or Application Errors) returned by AUSF services to the AMF and the 5GMM cause values sent to UEs.

### 4.2.2 Mapping between Nausf_UEAuthentication service causes on N12 and 5GMM causes

#### 4.2.2.1 General

This clause defines the mapping for the Nausf_UEAuthentication service (see 3GPP TS 29.509 [7]). It also contains the mapping in case of a 200 OK is returned but the Authentication has failed.

### 4.2.2.2 Mapping from HTTP to 5GMM causes values – Request rejected by AUSF

**Table .4.2.2-1: Mapping from HTTP to 5GMM cause values – Request rejected by AUSF**

| HTTP status code on N12 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 403 Forbidden | SERVING_NETWORK_NOT_AUTHORIZED | Cause #11 – PLMN not allowed<br>Cause #73 – Serving network not authorized<br>Cause #12 – Tracking area not allowed<br>Cause #15 – No suitable cells in tracking area<br><br>(NOTE1) |
| | AUTHENTICATION_REJECTED | N/A (NOTE 2) |
| | INVALID_HN_PUBLIC_KEY_IDENTIFIER | N/A (NOTE 2) |
| | INVALID_HN_PUBLIC_KEY_IDENTIFIER | N/A (NOTE 2) |
| 404 Not Found | CONTEXT_NOT_FOUND | N/A (NOTE 2) |
| | USER_NOT_FOUND | |
| 504 Gateway Timeout | UPSTREAM_SERVER_ERROR | N/A (NOTE 2) |
| | NETWORK_FAILURE | N/A (NOTE 2) |
| 500 Internal Server Error | AV_GENERATION_PROBLEM | N/A (NOTE 2) |
| 501 Not Implemented | UNSUPPORTED_PROTECTION_SCHEME | N/A (NOTE 2) |
| NOTE 1: .One of these 5GMM causes may be used.<br>NOTE 2: There is no corresponding mapping since the Authentication is rejected and the Authentication Reject message does not have a 5GMM cause. | | |

### 4.2.2.3 Mapping from HTTP to 5GMM cause values – Request accepted by AUSF

**Table .4.2.2.3-1: Mapping from HTTP to 5GMM cause values – Request accepted by AUSF**

| HTTP status code on N12 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 200 OK | The "authResult" in "ConfirmationDataResponse" is set to "AUTHENTICATION_FAILURE" (see clause 6.1.6.2.8 of 3GPP TS 29.509 [7]). | Cause #3 – Illegal UE |
| | The "authResult" in "EapSession" is set to "AUTHENTICATION_FAILURE" (see clause 6.1.6.2.8 of 3GPP TS 29.509 [7]). | Cause #3 – Illegal UE |

# 4.3 Mapping between Nsmf services causes and 5GMM causes

## 4.3.1 General

This clause defines the mapping performed by the AMF between HTTP responses (status codes and Protocol or Application Errors) returned by SMF services to the AMF and the 5GMM cause values sent to UEs.

The AMF may invoke SMF service operations upon receipt of:

- 5GSM messages (e.g. PDU Session establishment request) piggybacked in 5GMM UL NAS TRANSPORT message;
- 5GMM Service Request or Registration Request messages, to establish the user-plane resources of a PDU session.

A 5GMM cause is defined in:

- the DL NAS TRANSPORT message;
- the PDU session reactivation result error cause IE of a REGISTRATION ACCEPT or SERVICE ACCEPT message;
- the REGISTRATION REJECT or SERVICE REJECT message.

The AMF shall support mapping the status code and/or Protocol or Application Error received from the SMF to a 5GMM cause code, where 3GPP TS 24.501 [3] requires a 5GMM Cause to be included in the 5GMM message sent to the UE.

NOTE: REGISTRATION REJECT and SERVICE REJECT messages are rejected only due to 5GMM-specific reasons, i.e. not for problems specific to session management.

## 4.3.2 Mapping between Nsmf_PDUSession service causes on N11 and 5GMM causes

### 4.3.2.1 General

This clause defines the mapping for the Nsmf_PDUSession service (see 3GPP TS 29.502 [8]).

### 4.3.2.2 Mapping from HTTP to 5GMM Cause Values

**Table 4.3.2.2-1: Mapping from HTTP to 5GMM cause values – Request rejected by SMF**

| HTTP status code on N11 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 200 OK | The upCnxState IE is set to DEACTIVATED and the Cause IE set to "INSUFFICIENT_UP_RESOURCES" (see clause 5.2.2.3.2.2 of 3GPP TS 29.502 [8]) (NOTE 2) | #92 - Insufficient user-plane resources for the PDU session |
| 403 Forbidden | OUT_OF_LADN_SERVICE_AREA | #43 - LADN not available |
| 403 Forbidden | PRIORITIZED_SERVICES_ONLY | #28 - Restricted service area |
| 404 Not Found | CONTEXT_NOT_FOUND | N/A (NOTE 1) |
| | | |
| | | |
| | | |
| | | |
| 504 Gateway Timeout | NETWORK_FAILURE | N/A (NOTE 1) |
| NOTE 1: An N1 SM cause is included by the SMF in the HTTP error response in this case.<br>NOTE 2: This corresponds to the scenario where NG-RAN cannot establish user plane resources during the activation of the User Plane connection of a PDU session. | | |

# 4.4 Mapping between Nudm services causes and 5GMM causes

## 4.4.1 General

This clause defines the mapping performed by the AMF between HTTP responses (status codes and Protocol or Application Errors) returned by UDM services to AMF and the 5GMM cause values sent to UEs.

## 4.4.2 Mapping between Nudm_UEContextManagement service causes on N8 and 5GMM causes

### 4.4.2.1 General

This clause defines the mapping for the Nudm_UEContextManagement service is described in 3GPP TS 29.503 [5]).

### 4.4.2.1 Mapping from HTTP to NAS cause values – Request rejected by UDM

**Table 4.4.2.1-1: Mapping from HTTP to 5GMM cause values – Request rejected by UDM**

| HTTP status code on N8 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 403 Forbidden | UNKNOWN_5GS_SUBSCRIPTION | #27 – N1 mode not allowed |
| | ACCESS_NOT_ALLOWED | #15 "No suitable cells in tracking area", or<br>#12 "Tracking area not allowed" or<br>#72 – Non-3GPP access to 5GCN not allowed (NOTE) |
| | RAT_NOT ALLOWED | #15 "No suitable cells in tracking area", or<br>#13 "Roaming not allowed in this tracking area", or<br>#12 "Tracking area not allowed" (NOTE) |
| | NO_PS_SUBSCRIPTION | #7 – 5GS services not allowed |
| | ROAMING_NOT_ALLOWED | #11 "PLMN not allowed" or<br>#13 "Roaming not allowed in this tracking area",<br>(NOTE) |
| 404 Not Found | CONTEXT_NOT_FOUND | #9 "UE identity cannot be derived by the network" |
| 404 Not Found | USER_NOT_FOUND | #3 "Illegal UE" |
| 422 Unprocessable Entity | UNPROCESSABLE_REQUEST | #111 protocol error unspecified |
| NOTE: Any of those NAS Cause Code values may be sent to the UE, depending on operator's choice. | | |

## 4.4.3 Mapping between Nudm_SubscriberDataManagement service causes on N8 and 5GMM causes

### 4.4.3.1 General

This clause defines the mapping for the Nudm_SubscriberDataManagement service is described in 3GPP TS 29.503 [5]).

#### 4.4.3.2 Mapping from HTTP to NAS cause values – Request rejected by UDM due to N8 failure

**Table 4.4.x.2-1: Mapping from HTTP to 5GMM cause values – Request rejected by UDM due to N8 failures when consuming Nudm_SubscriberDataManagement service**

| HTTP status code on N8 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 404 Not Found | USER_NOT_FOUND<br>DATA_NOT_FOUND | #3 "Illegal UE"<br>#27 "N1 mode not allowed" |
| NOTE: Any of those NAS Cause Code values may be sent to the UE, depending on operator's choice. | | |

## 4.5 Mapping between Neir services causes and 5GMM causes

### 4.5.1 General

This clause defines the mapping performed by the AMF between HTTP responses (status code and Protocol or Application Errors) returned by the EIR to the AMF and the 5GMM cause values sent to UEs.

### 4.5.2 Mapping between N5g-eir_EquipmentIdentityCheck Service causes on N17 and 5GMM causes

#### 4.5.2.1 General

This clause defines the mapping for the N5g-eir_EquipmentIdentityCheck service (see 3GPP TS 29.511 [11]).

#### 4.5.2.2 Mapping from HTTP to NAS cause values – Request accepted by EIR

**Table 4.5.2.2-1: Mapping from HTTP to 5GMM cause values – Request accepted by EIR**

| HTTP Status on N17 | Protocol or Application information | 5GMM cause to UE |
|---|---|---|
| 200 OK | Equipment Status is set to "BLACKLISTED", which indicates that the equipment is in the prohibited list. | #6 "Illegal ME" |

## 4.6 Mapping between Nnssf services causes and 5GMM causes

### 4.6.1 General

This clause defines the mapping performed by the AMF between HTTP responses (Status Code and Protocol or Application Errors) returned by the NSSF to the AMF and the 5GMM cause values sent to UEs.

### 4.6.2 Mapping between Nnssf_NSSelection service causes on N22 and 5GMM causes

#### 4.6.2.1 General

This clause defines the mapping for the Nnssf_NSSelection service (see 3GPP TS 29.531 [4]).

#### 4.6.2.2 Mapping from HTTP to 5GMM Cause Values

**Table 4.6.2.2-1: Mapping from HTTP to 5GMM cause values – Request rejected by NSSF**

| HTTP status code on N22 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 403 Forbidden | SNSSAI_NOT_SUPPORTED | #62 – No Network Slices available |

## 4.7 Mapping between Nnsacf services causes and 5GMM causes

### 4.7.1 General

This clause defines the mapping performed by the AMF between HTTP responses (Status Code and Protocol or Application Errors) returned by the NSACF to the AMF and the 5GMM cause values sent to UEs.

### 4.7.2 Mapping between Nnsacf_NSAC service causes on N58 and 5GMM causes

#### 4.7.2.1 General

This clause defines the mapping for the Nnsacf_NSAC service (see 3GPP TS 29.536 [14]).

#### 4.7.2.2 Mapping from HTTP to 5GMM Cause Values

**Table 4.7.2.2-1: Mapping from HTTP to 5GMM cause values – Request rejected by NSACF**

| HTTP status code on N81 | Protocol or Application Error | 5GMM cause to UE |
|---|---|---|
| 403 Forbidden | ALL_SLICE_FAILED | #62 – No network slices available (NOTE) |
| NOTE: AMF shall reject the registration with 5GMM Cause #62 as specified in clause 4.6.2.5 of 3GPP TS 24.501 [3] if it receives application error of ALL_SLICE_FAILED and there is no S-NSSAI allowed for the UE. | | |

# 5 Mapping between 5GC interfaces causes and 5GSM Cause Codes by SMF

## 5.1 General

This clause describes the mapping performed by the SMF between HTTP responses (status Code and Protocol or Application Errors) returned by NF services contacted by the SMF and the corresponding 5GSM cause values sent to UEs (in5GSMmessages).

No mapping is required for the Nsmf_PDUSession service over N11 since the SMF includes the 5GSM cause in 5GSM messages sent in N1 SM Information (see 3GPP TS 29.502 [5]).

No mapping is required for the Nsmf_PDUSession service over N16 since the 5GSM cause is generated at the H-SMF and provided to the V-SMF in N1 SM Information (see 3GPP TS 29.502 [9]).

No mapping is required for the Nsmf_EventExposure service specified in 3GPP TS 29.508 [9]

# 5.2 Mapping between Npcf service causes on N7 and 5GSM causes

## 5.2.1 General

This clause defines the mapping performed by the SMF between HTTP responses (status code and Protocol or Application Errors) returned by the PCF to the SMF and the 5GSM causes sent to UEs.

## 5.2.2 Mapping between Npcf_SMPolicyControl Service causes on N7 and 5GSM causes

### 5.2.2.1 General

This clause defines the mapping for the Npcf_SMPolicyControl service (see 3GPPTS 29.512 [6]).

### 5.2.2.2 Mapping from HTTP to 5GSM cause values – Request rejected by PCF

**Table 5.2.2.2-1: Mapping from HTTP to 5GSM cause values – Request rejected by PCF**

| HTTP status code on N7 | Protocol or Application Error | 5GSM cause to UE |
|---|---|---|
| 400 Bad Request | USER_UNKNOWN | Cause #29 - User authentication or authorization failed |
| | ERROR_INITIAL_PARAMETERS | Cause #31 - Request rejected, unspecified |
| | ERROR_TRIGGER_EVENT | Cause #31 - Request rejected, unspecified |
| 403 Forbidden | ERROR_TRAFFIC_MAPPING_INFO_REJECTED | Cause #29 - User authentication or authorization failed |
| | POLICY_CONTEXT_DENIED | Cause #29 - User authentication or authorization failed (NOTE) |
| | VALIDATION_CONDITION_NOT_MET | Cause #29 - User authentication or authorization failed |
| | EXCEEDED_UE_SLICE_DATA_RATE | Cause #69 - Insufficient resources for specific slice |
| | EXCEEDED_SLICE_DATA_RATE | Cause #69 - Insufficient resources for specific slice |
| | EXCEEDED_GROUP_DATA_RATE | Cause #69 - Insufficient resources for specific slice |
| NOTE: Upon receiving the cause POLICY CONTEXT DENIED, the SMF may reject the request, or it may accept the request based on the local policy. | | |

# 5.3 Mapping between Nudm service causes on N10 and 5GSM causes

## 5.3.1 General

This clause defines the mapping performed by the SMF between HTTP responses (status codes and Protocol or Application Errors) returned by the UDM to the SMF and the 5GSM causes sent to UEs.

## 5.3.2 Mapping between Nudm_UEContextManagement service causes on N10 and 5GSM causes

### 5.3.2.1 General

This clause defines the mapping for the Nudm_UEContextManagement service (see 3GPP TS 29.503 [5]).

#### 5.3.2.2 Mapping from HTTP to 5GSM cause values – Request rejected by UDM due to N10 failures

**Table 5.3.2.2-1: Mapping from HTTP to 5GSM cause values – Request rejected by UDM due to N10 failures**

| HTTP status code on N10 | Protocol or Application Error | 5GSM cause to UE |
|---|---|---|
| 403 Forbidden | ROAMING_NOT_ALLOWED | #29 "User authentication or authorization failed" |
| | DNN_NOT ALLOWED | #27 "Missing or unknown DNN" or<br>#67 "Insufficient resources for specific slice and DNN"<br>#70 "Missing or unknown DNN in a slice"<br>(NOTE) |
| 404 Not Found | USER NOT FOUND | #29 "User authentication or authorization failed" |
| NOTE: Any of those NAS Cause Code values may be sent to the UE, depending on operator's choice. | | |

### 5.3.3 Mapping between Nudm_SubscriberDataManagement service causes on N10 and 5GSM causes

#### 5.3.3.1 General

This clause defines the mapping for the Nudm_SubscriberDataManagement service (see 3GPP TS 29.503 [5]).

#### 5.3.3.2 Mapping from HTTP to 5GSM cause values – Request rejected by UDM due to N10 failures

**Table 5.3.3.2-1: Mapping from HTTP to 5GSM cause values – Request rejected by UDM due to N10 failures when consuming Nudm_SubscriberDataManagement service**

| HTTP status code on N10 | Protocol or Application Error | 5GSM cause to UE |
|---|---|---|
| 404 Not Found | USER NOT FOUND<br>DATA_NOT_FOUND | #29 "User authentication or authorization failed"<br>#33 "Requested service option not subscribed" |
| NOTE: Any of those NAS Cause Code values may be sent to the UE, depending on operator's choice. | | |

## 5.4 Mapping between N4 causes and 5GSM causes

### 5.4.1 General

N4 error handling and cause codes are defined in 3GPP TS 29.244 [2].

The receipt of an N4 error may trigger error handling in SMF. This clause defines the mapping performed by the SMF between when an error received on N4 interface cannot be resolved by SMF and results in the SMF sending a 5GSM message to the UE.

## 5.4.2 Mapping from N4 to 5GSM cause values

**Table 5.4.2-1: Mapping from N4 to 5GSM cause values – Request rejected by SMF due to N4 failures**

| N4 Cause Code | | 5GSM cause to UE | |
|---|---|---|---|
| Cause value Decimal | Error description | Cause value | Error description |
| 64 | Request rejected (reason not specified) | #31 | Request rejected, unspecified |
| 74 | PFCP entity in congestion | #26<br>#38<br>#69<br><br>#67 | Insufficient resources<br>Network Failure<br>Insufficient resources for specific slice<br>Insufficient resources for specific slice and DNN |
| 77 | System failure | #31 | Request rejected, unspecified |

# 5.5 Mapping between Nnsacf services causes and 5GSM causes

## 5.5.1 General

This clause defines the mapping performed by the SMF between HTTP responses (Status Code and Protocol or Application Errors) returned by the NSACF to the SMF and the 5GSM cause values sent to UEs.

## 5.5.2 Mapping between Nnsacf_NSAC service causes on N81 and 5GSM causes

### 5.5.2.1 General

This clause defines the mapping for the Nnsacf_NSAC service (see 3GPP TS 29.536 [14]).

### 5.5.2.2 Mapping from HTTP to 5GSM Cause Values

**Table 5.X.2.2-1: Mapping from HTTP to 5GSM cause values – Request rejected by NSACF**

| HTTP status code on N81 | Protocol or Application Error | 5GSM cause to UE |
|---|---|---|
| 403 Forbidden | ALL_SLICE_FAILED | #69 –Insufficient resources for specific slice |

# Annex A (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2018-10 | CT4#86Bis | | | | | First version of this new TS including:<br>C4-187375 (Skeleton)<br>C4-187378 (N11 – 5GMM)<br>C4-187391 (N8 – 5GMM – Request rejected)<br>C4-187589 (Scope)<br>C4-187390 (N4 – 5GMM) | 0.1.0 |
| 2018-12 | CT4#87 | | | | | The following agreed documents are included:<br>C4-188418 (NSSF)<br>C4-188484 (Nsmf_PDUSession N11 and N16)<br>C4-188486 (Cause 5GMM)<br>C4-188609 (N4)<br>C4-188637 (N8 and 5GMM)<br>C4-188638 (EIR)<br>C4-188640 (N10 and 5GSM)<br>C4-188641 (N7)<br>C4-188642 (Nausf)<br>C4-188643 (N11 5GMM)<br>C4-188647 (Clean-up) (Conclusion from C4-188384 was wrongly reported to be included in 5GSM General Clause while it applied to 5GMM – corrected by Rapporteur).<br>C4-188648 (General 5GSM) | 0.2.0 |
| 2018-12 | CT#82 | CP-183104 | | | | Sent for information and approval | 1.0.0 |
| 2018-12 | CT#82 | CP-183104 | | | | Approved in CT#82 | 15.0.0 |
| 2019-03 | CT#83 | CP-190031 | 0001 | 2 | F | Corrections on cause mapping in clause 4 | 15.1.0 |
| 2019-03 | CT#83 | CP-190031 | 003 | 1 | F | Mapping between N11 causes and 5GMM causes | 15.1.0 |
| 2019-03 | CT#83 | CP-190031 | 0008 | 1 | F | Remove EN's in 5.2.2.2 | 15.1.0 |
| 2019-03 | CT#83 | CP-190031 | 0004 | 2 | F | Corrections to 5GMM and 5GSM causes mappings | 15.1.0 |
| 2019-06 | CT#84 | CP-191037 | 0012 | - | F | Cause mapping updates | 15.2.0 |
| 2019-06 | CT#84 | CP-191037 | 0013 | 1 | F | Correction on cause mapping regarding no subscription | 15.2.0 |
| 2019-06 | CT#84 | CP-191037 | 0014 | 2 | F | Essential Correction on Error mapping for NSSF Services | 15.2.0 |
| 2019-06 | CT#84 | CP-191037 | 0015 | 1 | F | Essential Correction on mapping for N4 to SBI | 15.2.0 |
| 2019-06 | CT#84 | CP-191037 | 0016 | 1 | F | Cause mapping corrections | 15.2.0 |
| 2019-06 | CT#84 | CP-191037 | 0017 | 1 | F | Add a mapping for N10 | 15.2.0 |
| 2019-12 | CT#86 | CP-193036 | 0019 | 2 | F | Add new mapping towards Cause #62 (No Network Slices available) | 16.0.0 |
| 2020-06 | CT#88e | CP-201034 | 0020 | 1 | F | Correction of the mapping between PCF service causes and 5GSM causes | 16.1.0 |
| 2021-03 | CT#91e | CP-210027 | 0022 | 1 | D | Use of inclusive terminology | 17.0.0 |
| 2021-03 | CT#91e | CP-210028 | 0023 | 1 | F | ERROR CONFLICTING REQUEST | 17.0.0 |
| 2021-06 | CT#92e | CP-211146 | 0024 | | F | Editorial Correction | 17.1.0 |
| 2021-09 | CT#93e | CP-212230 | 0025 | 1 | B | Cause Mapping for NSAC | 17.2.0 |
| 2021-12 | CT#94e | CP-213086 | 0026 | - | F | Mapping for the cause from Nudm_SubscriberDataManagement | 17.3.0 |
| 2023-12 | CT#102 | CP-233064 | 0028 | 1 | F | Cause mapping for EXCEEDED_UE_SLICE_DATA_RATE and EXCEEDED_SLICE_DATA_RATE | 17.4.0 |
| 2023-12 | CT#102 | CP-233047 | 0027 | - | B | Cause mapping for EXCEEDED_GROUP_DATA_RATE | 18.0.0 |
| 2024-03 | CT#103 | CP+240042 | 0029 | 1 | B | Cause Mapping for Network Slice Operations | 18.1.0 |
| 2025-10 | - | - | - | - | - | Update to Rel-19 version (MCC) | **19.0.0** |

# History

| Document history | | |
|---|---|---|
| V19.0.0 | October 2025 | Publication |
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